Note 9 ARMA Models

1 ARMA Models

Recall: A MA(q) model is

$$X_t = \mu + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \cdots - \theta_q Z_{t-q},$$

and an AR(p) model is

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots \phi_p X_{t-p} = C + Z_t$$

Here $\{Z_t\}$ is (Gaussian) white noise with mean zero and variance σ_Z^2.

If a time series $\{X_t\}$ is generated from the model

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = C + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \cdots - \theta_q Z_{t-q},$$

it is called an **AutoRegressive Moving-Average** model with AR order p and MA order q, denoted as ARMA(p, q). Here $\{Z_t\}$ is (Gaussian) white noise with mean zero and variance σ_Z^2.

Note: When $p = 0$, an ARMA(p, q) model is a MA(q) model; when $q = 0$, an ARMA(p, q) model is an AR(p) model.

- Not all ARMA(p, q) models are stationary.
- For a stationary ARMA model, the constant C is not arbitrary.

- Using the backshift operator, an ARMA(p, q) model can be written as

2 Parameter Redundancy

Let $\{X_t\}$ be a white noise model, that is,

$$X_t = Z_t. \tag{1}$$

Consider another model,

$$X_t - 0.5X_{t-1} = Z_t - 0.5Z_{t-1} \tag{2}$$

In an ARMA(p, q) model, we require it is in its simplest form. That is, let the AR and MA polynomial be

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$$

and

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q,$$

we require $\phi(B)$ and $\theta(B)$ have no common factors. Another way to phrase is, we require the equations $\phi(B) = 0$ and $\theta(B) = 0$ have no common roots.

Example 2.1. Are the ARMA model below in its simplest form? If not, find its simplest form.

$$X_t = 2 - 1.5X_{t-1} + X_{t-2} + Z_t + 1.75Z_{t-1} - 0.5Z_{t-2}$$

3 Causality

Recall: If a time series $\{X_t\}$ can be expressed in the following form

$$X_t = C + \sum_{i=0}^{\infty} \psi_i Z_{t-i},$$

with $\sum_{i=0}^{\infty} |\psi_i| < \infty$, then $\{X_t\}$ is said to be causal.

An ARMA(p, q) model is causal when all the roots of the AR polynomial

$$\phi(B) = 0$$

lie outside the unit circle.

4 Invertibility

Recall: If a time series $\{X_t\}$ can be expressed in the following form

$$X_t - \sum_{i=1}^{\infty} \pi_i X_{t-i} = C + Z_t,$$

with $\sum_{i=1}^{\infty} |\pi_i| < \infty$, then $\{X_t\}$ is said to be invertible.

An ARMA(p, q) model is causal when all the roots of the MA polynomial

$$\theta(B) = 0$$

lie outside the unit circle.

Example 4.1. Consider the ARMA model

$$X_t = 2 - 1.5X_{t-1} + X_{t-2} + Z_t + 1.75Z_{t-1} - 0.5Z_{t-2}$$

Is it causal? Is is invertible?

4.1 ψ weights and π weights

Example 4.2. We in example 4.1 the ARMA model with the following the simplest form is both causal and invertible

$$X_t - 0.5X_{t-1} = \frac{2}{3} + Z_t - 0.25Z_{t-1}$$

Find its ψ weights and π weights.

5 Model Identification

5.1 ACF

5.2 PACF

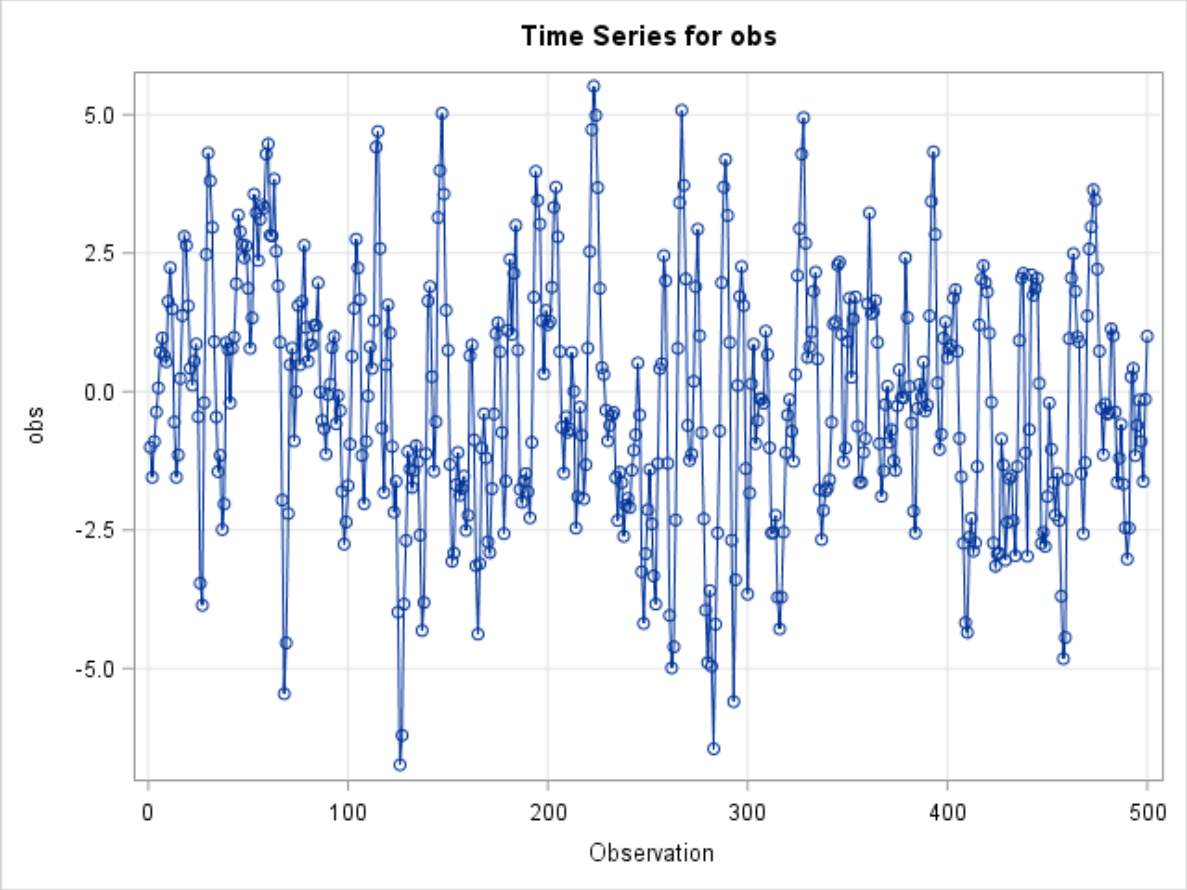
5.3 Summary

5.4 An ad-hoc way

We are given a stationary series $\{X_t\}$. How to choose a proper ARMA(p, q) model for $\{X_t\}$?

1. Start with an AR(1) model to $\{X_t\}$ and check the sample ACF of the residuals.
   - If the sample ACF of the residuals cuts off at q (q rarely exceeds 3 for model parsimony), tentatively try an ARMA(1,q) model for $\{X_t\}$;
   - Otherwise, fit an AR(2) model to $\{X_t\}$, check the sample ACF of the residuals, and identify an order for the MA part.
2. Estimate the parameters (MLE is recommended), check the p-values, make modifications if necessary.

Example 5.1. Choose an ARMA model for the series below.



5.5 Select Order by AIC/SBC

6 Estimation

Maximum likelihood method, unconditional least square method, and conditional least square method all can be used for ARMA models.

7 Model Diagnostics

8 Forecast

8.1 Point Estimate

8.2 Prediction Interval

Example 8.1. In example 4.1, we have showed the following ARMA model is both causal and invertible.

$$X_t - 0.5X_{t-1} = \frac{2}{3} + Z_t - 0.25Z_{t-1}$$

If $X_{99} = 3, X_{100} = 2$, forecast X_{101} and find its 95% confidence limit.